

Laurent Colbois

POSTDOCTORAL RESEARCHER · VISION-LANGUAGE MODELS FOR FORENSIC BIOMETRICS

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“To create a positive impact in local communities while following and supporting sustainable practices.”

Experience

Idiap Research Institute / University of Lausanne | School of Criminal Justice

Martigny, Switzerland

PHD STUDENT - DEEPFAKES, BIOMETRICS, FORENSIC SCIENCE

Jan. 2021 - Apr. 2025

- Developed systems for detection of AI-generated images, with a focus on fake passport photos
- Evaluated the vulnerability of face recognition systems to deepfakes
- Presented outreach talks on AI and deepfakes to the forensic science community and to police forces
- Supervised two master theses on score calibration for forensic face recognition
- Published several research code packages targeting experiments reproducibility

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN

Jul. - Dec. 2020

- Created and published a synthetic face recognition dataset using generative AI (GANs)
- Evaluated the applicability of this dataset for benchmarking face recognition systems

Logitech

Lausanne, Switzerland

RESEARCH INTERN

Feb. - Aug. 2019

- Developed an automated audio scene classification model for auto-equalization of a loudspeaker
- Implemented a real-time demo running on a microcontroller

Education

EPFL

Lausanne, Switzerland

M.SC. IN COMPUTATIONAL SCIENCE & ENGINEERING

2017 - 2020

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing
- GPA: 5.72/6

EPFL

Lausanne, Switzerland

B.SC. IN PHYSICS

2013 - 2016

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing
- GPA: 5.51/6

Publications

Year **Conference Paper**, Add your publications here using the cvhonor format

Skills

Domains Biometrics, Machine learning, Deep learning, Computer vision, Audio processing, Numerical analysis, Reproducible research, Scientific communication

Programming Python, PyTorch, Scikit-Learn, Pandas, Unix, Git, Conda, Snakemake, C++

Languages French (Mother tongue), English (Professional proficiency), German (B2)